Chapter 1

1.3 Complex Numbers

Definition of complex number

A complex number is a combination of a real number and imaginary number.
Standard form:

$$a + bi$$

where *a* and *b* are real numbers and *i* is the imaginary unit.

$$i = \sqrt{-1}$$
$$i^2 = -1$$

Example 1

Evaluate the following.

a) i^{15} b) i^{90} c) i^{35}

Solution

a) $i^{15} = i^{14} \cdot i = (i^2)^7 \cdot i = (-1)^7 \cdot i = -i$
b) $i^{90} = (i^2)^{45} = (-1)^{45} = -1$
c) $i^{35} =$

Equality of Complex Numbers

If $a + bi = c + di$, then $a = c$ and $b = d$.

Operation of Complex Numbers

a) Addition

$$(a + bi) + (x + yi) = a + x + bi + yi$$
$$= (a + x) + (b + y)i$$

b) Subtraction

$$(a + bi) - (x + yi) = a - x + bi - yi$$
$$= (a - x) + (b - y)i$$

c) Multiplication

$$(a + bi)(x + yi) = ax + ayi + bxi + byi^2$$
$$= ax + by(-1) + ayi + bxi$$
$$= (ax - by) + (ay + bx)i$$

Example 2

Solve the following complex numbers.

a) $(3+5i)+(10-2i)$

b) $(2-6i)-(8+4i)$

c) $(2+3i)(4-2i)$

Solution

a) $(3+5i)+(10-2i) = 3+5i+10-2i = 13+3i$

b) $(2-6i)-(8+4i) = 2-6i-8-4i = -6-10i$

c) $$\begin{aligned}(2+3i)(4-2i) &= 2(4-2i)+3i(4-2i)\\ &= 8-4i+12i-6i^2\\ &= 8-4i+12i-6(-1)\\ &= 8-4i+12i+6\\ &= 14+8i\end{aligned}$$

Magnitude of Complex Numbers, $|z|$

If $z = a+bi$, then $$\begin{aligned}|z| &= |a+bi|\\ &= \sqrt{(a)^2+(b)^2}\end{aligned}$$

Example 3

Find the magnitude of the following complex numbers.

a) $4+3i$
b) $(2+i)(1-3i)$

Solution

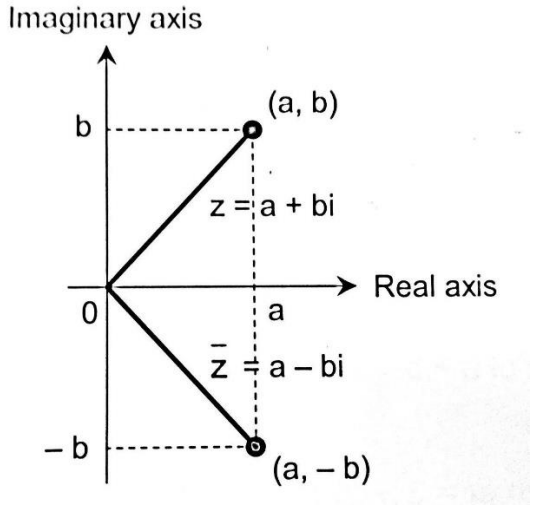
a) $|4+3i| = \sqrt{(4)^2+(3)^2} = \sqrt{25} = 5$

b) $$\begin{aligned}(2+i)(1-3i) &= 2(1-3i)+i(1-3i)\\ &= 2-6i+i-3i^2\\ &= 2-6i+i-3(-1)\\ &= 2-6i+i+3\\ &= 5-5i\end{aligned}$$

$i^2 = -1$

$\therefore\ |5-5i| = \sqrt{(5)^2+(-5)^2} = \sqrt{50} = 5\sqrt{2}$

Conjugate of Complex Numbers, $\bar{z}$



If $\boldsymbol{z = a + bi}$, then its conjugate $\boldsymbol{\bar{z} = a - bi}$

Example 4

State the conjugate of the following complex numbers.

a) $z = 2 + i$ b) $z = -3 - 2i$ c) $-3i + 5$

Solution

a) $\bar{z} = 2 - i$
b) $\bar{z} = -3 + 2i$
c) $\bar{z} = 3i + 5$

Application of Conjugate: Division of Complex Numbers

$$\frac{\text{Numerator}}{\text{Denominator}} \times \left[\frac{\text{Conjugate of Denominator}}{\text{Conjugate of Denominator}}\right]$$

Example 5

Express $\frac{3-4i}{5+3i}$ in the standard form $a+bi$.

Solution

$$\frac{3-4i}{5+3i}=\frac{3-4i}{5+3i}\times\frac{5-3i}{5-3i} \qquad \frac{\text{Numerator}}{\text{Denominator}}\times\left[\frac{\text{Conjugate of Denominator}}{\text{Conjugate of Denominator}}\right]$$

$$=\frac{(3-4i)(5-3i)}{(5+3i)(5-3i)}$$

$$=\frac{3(5-3i)-4i(5-3i)}{5(5-3i)+3i(5-3i)}$$

$$=\frac{15-9i-20i+12i^2}{25-15i+15i-9i^2}$$

$$=\frac{15-29i+12(-1)}{25-9(-1)} \qquad \boxed{i^2=-1}$$

$$=\frac{15-29i-12}{25+9}$$

$$=\frac{3-29i}{34}$$

$$=\frac{3}{34}-\frac{29}{34}i \qquad \textbf{(Standard form)}$$

Candidate's mistake:

$$\frac{3-4i}{5+3i}\times\frac{-5+3i}{-5+3i} \qquad \boxed{\textbf{Wrong Conjugate}}$$

Example 6

Given $z = -\sqrt{2} + 2i$. Write the expression $\frac{1}{i^{19}z}$ in standard form $a + bi$.

Solution

Firstly solve $i^{19} = i^{18} \cdot i = (i^2)^9 \cdot i = (-1)^9 \cdot i = -i$

$$\frac{1}{i^{19}z} = \frac{1}{-iz}$$
$$= \frac{1}{-i(-\sqrt{2} + 2i)}$$
$$= \frac{1}{\sqrt{2}\,i - 2i^2}$$
$$= \frac{1}{\sqrt{2}\,i - 2(-1)} \qquad \boxed{i^2 = -1}$$
$$= \frac{1}{\sqrt{2}\,i + 2}$$
$$= \frac{1}{\sqrt{2}\,i + 2} \times \frac{-\sqrt{2}\,i + 2}{-\sqrt{2}\,i + 2} \qquad \frac{\text{Numerator}}{\text{Denominator}} \times \left[\frac{\text{Conjugate of Denominator}}{\text{Conjugate of Denominator}}\right]$$
$$= \frac{-\sqrt{2}\,i + 2}{(\sqrt{2}\,i + 2)(-\sqrt{2}\,i + 2)}$$
$$= \frac{-\sqrt{2}\,i + 2}{-2i^2 + 2\sqrt{2}i - 2\sqrt{2}i + 4}$$
$$= \frac{-\sqrt{2}\,i + 2}{-2(-1) + 4}$$
$$= \frac{-\sqrt{2}\,i + 2}{2 + 4}$$
$$= \frac{-\sqrt{2}\,i + 2}{6}$$
$$= \frac{-\sqrt{2}\,i}{6} + \frac{2}{6}$$
$$= \frac{1}{3} - \frac{\sqrt{2}}{6}i \qquad \textbf{(Standard form)}$$

Candidate's mistake:

$$\frac{1}{\sqrt{2}\,i + 2} \times \frac{\sqrt{2}\,i - 2}{\sqrt{2}\,i - 2} \qquad \boxed{\textbf{Wrong Conjugate}}$$